

LECTURE #3

Econometrics I

OLS PROPERTIES

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In the previous lecture #2

- ▶ We discussed the types of data analyzed in econometrics.
- ▶ We defined the **simple linear regression model**

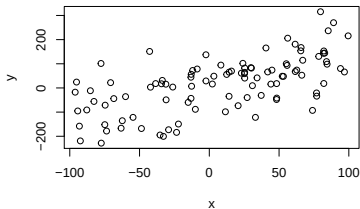
$$y = \beta_0 + \beta_1 x + u.$$

- ▶ From $\mathbb{E}(u) = 0$ and the **zero conditional mean assumption** $\mathbb{E}(u|x) = 0$, we got $\text{Cov}(x, u) = \mathbb{E}(xu) = 0$.
- ▶ We derived the **OLS estimators** (MM or LS approach):

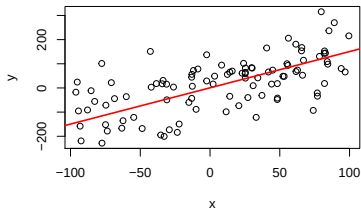
$$\hat{\beta}_1^{OLS} = \frac{\sum_{i=1}^n x_i(y_i - \bar{y})}{\sum_{i=1}^n x_i(x_i - \bar{x})} \quad \text{and} \quad \hat{\beta}_0^{OLS} = \bar{y} - \hat{\beta}_1 \bar{x}.$$

- ▶ Alternatively: $\hat{\beta}_1^{OLS} = \frac{\sum_{i=1}^n y_i(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \boxed{\frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}}.$
- ▶ Readings for lecture #3:
 - ▶ Chapter 2: 2.3, 2.5, **2.6 (mandatory for/after seminars)**.

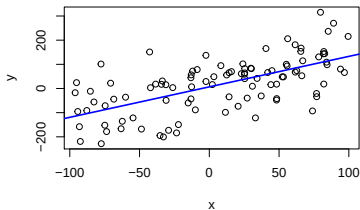
(a) $y = 0.5 + 1.5x + u$, $n = 100$



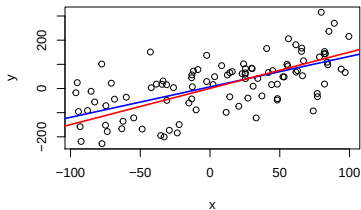
(b) PRF: $\mathbb{E}(y|x) = 0.5 + 1.5x$



(c) SRF (OLS RL): $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$



(d) SRF $\neq$ PRF



Outline

Basic OLS properties

Expected values and variances of the OLS estimators

Unbiasedness

Variance

Regression through the origin

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Algebraic properties of the OLS statistics

1. Sum (sample mean) of the OLS residuals is zero:

$$\sum_{i=1}^n \hat{u}_i = 0.$$

2. Sample covariance between the explanatory variables and the OLS residuals is zero:

$$\sum_{i=1}^n x_i \hat{u}_i = 0.$$

3. Observed value of y can be split into two uncorrelated parts, such that

$$y_i = \hat{y}_i + \hat{u}_i \quad \text{and} \quad \sum_{i=1}^n \hat{y}_i \hat{u}_i = 0.$$

4. Sample means of the observed y and fitted values are equal:

$$\bar{y} = \bar{\hat{y}} \quad \text{or alternatively} \quad \sum_{i=1}^n y_i = \sum_{i=1}^n \hat{y}_i.$$

5. Point $(\bar{x}, \bar{y})$ is always on the OLS regression line.

Algebraic properties of the OLS statistics: Proofs

- 1.,2. First two are given by the MM and LS derivations of the estimators; in fact, $\hat{\beta}_1$ and $\hat{\beta}_0$ are chosen to make them hold.
3. Observed value of y can be split into two uncorrelated parts:

$$\boxed{\hat{y}_i + \hat{u}_i} = \hat{y}_i + (y_i - \hat{y}_i) = \boxed{y_i},$$

$$\boxed{\sum_{i=1}^n \hat{y}_i \hat{u}_i} = \sum (\hat{\beta}_0 + \hat{\beta}_1 x_i) \hat{u}_i = \hat{\beta}_0 \sum \hat{u}_i + \hat{\beta}_1 \sum x_i \hat{u}_i = \boxed{0}.$$

4. Sample means of the observed and fitted values are equal:

$$\boxed{\sum y_i} = \sum (\hat{y}_i + \hat{u}_i) = \sum \hat{y}_i + \sum \hat{u}_i = \boxed{\sum \hat{y}_i}.$$

5. Point $(\bar{x}, \bar{y})$ is always on the OLS regression line:

$$\boxed{\hat{y}} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x} = \bar{y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 \bar{x} = \boxed{\bar{y}}.$$

Various 'sums of squares'

- ▶ Total sum of squares (SST)

$$SST \equiv \sum_{i=1}^n (y_i - \bar{y})^2.$$

- ▶ Explained sum of squares (SSE)

$$SSE \equiv \sum_{i=1}^n (\hat{y}_i - \bar{y})^2.$$

- ▶ Residual sum of squares (SSR)

$$SSR \equiv \sum_{i=1}^n \hat{u}_i^2.$$

- ▶ It holds that

$$SST = SSE + SSR.$$

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- ▶ We need to use a little trick of 'adding zero' to the sum:

$$\begin{aligned} SST &= \sum (y_i - \bar{y})^2 = \sum ((y_i - \hat{y}_i) + (\hat{y}_i - \bar{y}))^2 = \\ &= \sum (y_i - \hat{y}_i)^2 + 2 \sum \underbrace{(y_i - \hat{y}_i)}_{\hat{u}_i} (\hat{y}_i - \bar{y}) + \sum (\hat{y}_i - \bar{y})^2 = \\ &= SSE + 2 \sum \hat{u}_i (\hat{y}_i - \bar{y}) + SSR. \end{aligned}$$

- ▶ We thus need to show that $\sum \hat{u}_i (\hat{y}_i - \bar{y}) = 0$:
 - ▶ in the algebraic properties of OLS, we have already shown that $\sum \hat{u}_i \hat{y}_i = 0$,
 - ▶ and also $\sum \hat{u}_i = 0$ so that $\sum \hat{u}_i \bar{y} = \bar{y} \sum \hat{u}_i = \bar{y} \cdot 0 = 0$.

Goodness-of-fit

- ▶ We need to measure how well our model (or now specifically variable x) explains the variation in y .
- ▶ **Coefficient of determination**, or **R-squared**, is defined

$$R^2 \equiv \frac{SSE}{SST} = 1 - \frac{SSR}{SST}.$$

- ▶ R^2 can be interpreted (for the simple regression) as a fraction of the sample variation in y explained by x .
- ▶ R^2 ranges between $\langle 0, 1 \rangle$ and is sometimes reported in percentages.
- ▶ R^2 cannot decrease after adding an independent variable.
- ▶ Threshold values differ across disciplines and even across fields of economics and finance (usually also data-type dependent).

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Unbiasedness of OLS

Simple linear regression (SLR) assumptions:

- ▶ **SLR.1 Linear in parameters:** We have the population model

$$y = \beta_0 + \beta_1 x + u,$$

where β_0 is the population intercept and β_1 is the population slope parameter. The inclusion of β_0 implies $\mathbb{E}(u) = 0$.

- ▶ **SLR.2 Random sampling:** We have a random sample of size n following the population model.
- ▶ **SLR.3 Sample variation in the explanatory variable:**
The sample outcomes on x are not all the same value.
- ▶ **SLR.4 Zero conditional mean:** The error u has an expected value of zero given any value of the explanatory variable, i.e., $\mathbb{E}(u|x) = 0$. It implies $\text{Cov}(x, u) = 0$ and also $\mathbb{E}(u) = 0$.

Unbiasedness of the OLS estimators

Assuming SLR.1 through SLR.4, $\mathbb{E}(\hat{\beta}_0^{OLS}) = \beta_0$ and $\mathbb{E}(\hat{\beta}_1^{OLS}) = \beta_1$ for any values of β_0 and β_1 .

In other words, $\hat{\beta}_0^{OLS}$ **is unbiased for β_0** and $\hat{\beta}_1^{OLS}$ **is unbiased for β_1** .

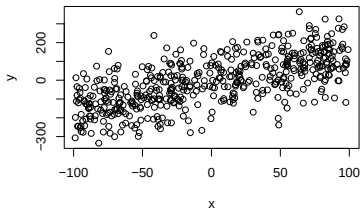
Unbiasedness of the OLS estimator $\hat{\beta}_1$: Proof

- ▶ We first need to rewrite the OLS estimator (see lecture #2 Appendix) as

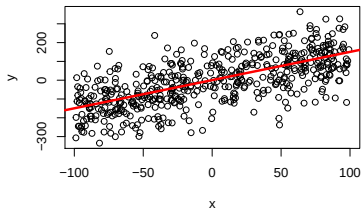
$$\begin{aligned}\boxed{\hat{\beta}_1} &\stackrel{SLR.3}{=} \frac{\sum y_i(x_i - \bar{x})}{\sum (x_i - \bar{x})^2} \stackrel{SLR.1}{=} \frac{\sum (\beta_0 + \beta_1 x_i + u_i)(x_i - \bar{x})}{\sum (x_i - \bar{x})^2} = \\ &= \beta_0 \frac{\sum (x_i - \bar{x})}{\sum (x_i - \bar{x})^2} + \beta_1 \frac{\sum x_i(x_i - \bar{x})}{\sum (x_i - \bar{x})^2} + \frac{\sum u_i(x_i - \bar{x})}{\sum (x_i - \bar{x})^2} = \\ &= 0 + \boxed{\beta_1 + \frac{\sum u_i(x_i - \bar{x})}{\sum (x_i - \bar{x})^2}}.\end{aligned}$$

- ▶ OLS estimator $\hat{\beta}_1$ can thus be expressed as the true β_1 plus an additional term, a linear combination of errors $\{u_1, u_2, \dots, u_n\}$.
- ▶ This is where its stochasticity comes from!
- ▶ We now need to show its expected value.

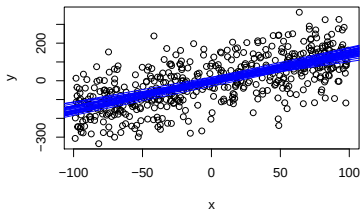
(a) $y = 0.5 + 1.5x + u$, $nn = 500$



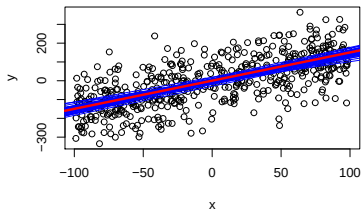
(b) PRF: $\mathbb{E}(y|x) = 0.5 + 1.5x$



(c) $30 \times \text{SRF}$: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$, $n = 100$



(d) $\mathbb{E}(\hat{\beta}_0) = 0.5$ and $\mathbb{E}(\hat{\beta}_1) = 1.5$



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Variance of the OLS estimators

Additional assumption:

- ▶ **SLR.5 Homoskedasticity:** The error u has the same variance given any value of the explanatory variable, i.e.,

$$\text{Var}(u|x) = \sigma^2.$$

- ▶ Homoskedasticity vs. heteroskedasticity.
- ▶ SLR.5 implies $\text{Var}(y|x) = \sigma^2$.

Variance of the OLS estimators

- ▶ It is also crucial to know how far we can expect $\hat{\beta}_1$ to be away from β_1 on average, i.e., how precise the estimator is.
- ▶ Assuming SLR.1 through SLR.5,

$$\begin{aligned}\text{Var}(\hat{\beta}_1) &= \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}, \\ \left(\text{Var}(\hat{\beta}_0) &= \frac{\sigma^2 \sum_{i=1}^n x_i^2}{n \sum_{i=1}^n (x_i - \bar{x})^2} \right).\end{aligned}$$

Variance of the OLS estimator $\hat{\beta}_1$: Derivation

- ▶ We will use the rewritten estimator $\hat{\beta}_1 = \beta_1 + \frac{\sum u_i(x_i - \bar{x})}{\sum (x_i - \bar{x})^2}$ as a starting point.
- ▶ As the variance of a parameter (constant) is zero, we can write

$$\begin{aligned}\boxed{\text{Var}(\hat{\beta}_1)} &= \text{Var}\left(\frac{\sum u_i(x_i - \bar{x})}{\sum (x_i - \bar{x})^2}\right) \stackrel{\text{SLR.4}}{=} \text{SLR.2} \\&= \frac{1}{(\sum (x_i - \bar{x})^2)^2} \text{Var}\left(\sum u_i(x_i - \bar{x})\right) \stackrel{\text{SLR.2}}{=} \\&= \frac{1}{(\sum (x_i - \bar{x})^2)^2} \sum \text{Var}(u_i(x_i - \bar{x})) = \\&= \frac{1}{(\sum (x_i - \bar{x})^2)^2} \sum (x_i - \bar{x})^2 \text{Var}(u_i) \stackrel{\text{SLR.5}}{=} \\&= \sigma^2 \frac{\sum (x_i - \bar{x})^2}{(\sum (x_i - \bar{x})^2)^2} = \boxed{\frac{\sigma^2}{\sum (x_i - \bar{x})^2}}.\end{aligned}$$

Estimating the error variance

- ▶ σ^2 from the previous slides is not observed and hardly ever known $\Rightarrow$ it also needs to be estimated from data.
- ▶ One would want to use $\frac{1}{n} \sum u_i^2$ as an estimator of σ^2 because

$$\boxed{\text{Var}(u|x)} \stackrel{SLR.5}{=} \sigma^2 = \mathbb{E}(u^2|x) - (\mathbb{E}(u|x))^2 \stackrel{SLR.4}{=} \boxed{\mathbb{E}(u^2|x)},$$

which is, unfortunately, not feasible.

- ▶ Errors u (unknown) vs. residuals $\hat{u}$ (after estimation).
- $\Rightarrow$ Under SLR.1 through SLR.5, **the unbiased estimator of σ^2** is given as

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n \hat{u}_i^2.$$

- ▶ $n - 2$ because we lose two degrees of freedom due to two restrictions on residuals:

$$\sum_{i=1}^n \hat{u}_i = 0, \quad \sum_{i=1}^n x_i \hat{u}_i = 0.$$

Estimating the error variance

- ▶ $\hat{\sigma}$ is called the **standard error of the regression**.
- ▶ **Standard error of $\hat{\beta}_1$** is then

$$se(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}}.$$

- ▶ $se(\hat{\beta}_1)$ is necessary to construct test statistics and confidence intervals.
- ▶ Note: you can consult the attached R code (not mandatory) that compares the theoretical $sd(\beta_1)$ and estimated $se(\hat{\beta}_1)$ in simulations.

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(readings: 2.6)

- In rare cases, assuming $\beta_0 = 0$, we are interested in a model

$$y = \beta_1 x + u.$$

- Both the method of moments and the least squares estimation via minimizing $SSR = \sum_{i=1}^n (y_i - \tilde{\beta}_1 x_i)^2$ lead to

$$\sum_{i=1}^n (x_i(y_i - \tilde{\beta}_1 x_i)) = 0. \quad (1)$$

- Solving Eq. 1 leads to

$$\tilde{\beta}_1 = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}.$$

- If $\beta_0 \neq 0$ then $\tilde{\beta}_1$ is biased $\Rightarrow$ not often used in practice.
- Mind the difference between the definition of R^2 of a standard regression and a regression through the origin!

Seminars and the next lecture

- ▶ Seminars:
 - ▶ interpretation of estimates and causality recap,
 - ▶ SLR.5 (homoskedasticity) violation,
 - ▶ regression through the origin: consequences,
 - ▶ computer exercise with simulated data (BYOD?).
- ▶ Next lecture #4:
 - ▶ multiple regression model and OLS,
 - ▶ expected value of the OLS estimators
 - ▶ unbiasedness,
 - ▶ irrelevant variables,
 - ▶ omitted variables,
 - ▶ variance of the OLS estimators (multicollinearity).
- ▶ Readings for lecture #4:
 - ▶ Chapter 3: 3.1–3.4, 3.6 (**3.1, esp. pg. 70, and 3.4, sections ‘Multicollinearity’ and ‘Misspecified models’ mandatory**).